



Sr. Data Engineer

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Professional Summary:

Over 10+ years of experience as an AWS Data Engineer, Big Data Engineer, ETL Developer, and Data Warehouse Developer, applying data mapping, validation, and statistical analysis to translate business requirements into actionable insights.

Proficient in AWS Services, Cloudera, Hadoop Ecosystem, Spark (PySpark/Scala), Databricks, MapReduce, Tez, Hive, Redshift, Snowflake, relational databases, and visualization tools like Tableau.

Experienced in AWS services like AWS Glue, AWS Lambda, Amazon Redshift, AWS CloudWatch, AWS Security Hub, Amazon Kinesis, AWS EMR, AWS Step Functions, Amazon DynamoDB, and AWS Lake Formation.

Worked extensively with Big Data technologies, including HDFS, MapReduce, YARN, Pig, Hive, Sqoop, Oozie, Zookeeper, Kafka, Cassandra (CDL), and Teradata, for a robust and scalable big data infrastructure.

Developed scalable distributed data processing solutions using Spark Streaming, Kafka, and Amazon Kinesis, enabling real-time data ingestion and processing for enterprise analytics.

Integrated on-premises and cloud-based data sources with AWS Glue, applying data transformations to manage seamless data movement to Snowflake.

Established a scalable data ecosystem by integrating AWS Glue with Amazon S3, Amazon Redshift, AWS Lambda, AWS EMR, and Amazon Athena.

Designed and implemented end-to-end data integration pipelines with AWS Glue, optimizing ETL/ELT workflows for efficient data movement and transformation.

Built data pipelines and transformations using Amazon S3, Amazon Redshift, AWS Lambda, and PySpark with AWS EMR, ensuring structured data processing.

Developed ETL transformations with PySpark/Spark-SQL in AWS Glue, automating data validation and transformation workflows through Python scripts in AWS Step Functions.

Engineered Delta Lake architecture as a curated data layer in Amazon S3, implementing encryption and role-based access control (RBAC) for security and compliance.

Experienced working with various serialized file formats, including Avro, Parquet, ORC, JSON, and XML, optimizing data storage, processing efficiency, and interoperability across big data and cloud platforms.

Applied advanced data quality checks, data cleansing, and validation techniques, enhancing data integrity, consistency, and accuracy across multiple pipelines.

Collaborated with AWS Step Functions administrators and DevOps engineers for identifying and resolving automation and data processing pipeline issues.

Utilized Python and Scala to build scalable data processing frameworks, automate ETL processes, and develop custom data transformations for big data applications.

Designed, implemented, and optimized database structures, including complex queries, stored procedures, and triggers to ensure high performance and scalability of Oracle-based data systems.

Built and maintained scalable data pipelines, managed data flows, and performed complex data transformations for large datasets in data warehousing environments.

Leveraged AWS services like S3, Lambda, and Redshift to build cloud-based data architectures and optimize data processing workflows.

and maintained data warehouses, ensured data consistency and accuracy, and utilized data visualization tools like Tableau and Power BI to present actionable business insights.

Applied CMMI practices managing data engineering projects, ensuring high-quality deliverables, consistent documentation, and adherence to process standards.

Designed Hadoop-RDBMS integration programs by leveraging Sqoop for smooth data migration between various structured and semi-structured data environments.

Built and optimized Spark jobs with Scala, PySpark, and Spark SQL, executing Hive scripts via Hive on Spark to enable big data processing and analysis.

Applied MapReduce techniques for achieving parallelized computation and distributed data processing, optimizing large-scale computations.

Developed and optimized data models and schemas with Apache Hive, Apache HBase, and Snowflake, ensuring efficient data retrieval, storage, and analytics.

Designed high-throughput data ingestion pipelines using Apache Kafka, enabling real-time data streaming for business intelligence applications.

Integrated AWS CloudWatch, AWS Security Hub, and Prometheus to create real-time performance monitoring dashboards, ensuring data pipeline health and proactive incident management.

Developed and automated ETL/ELT workflows with Apache Spark, Apache Beam, and Apache Airflow, ensuring seamless data movement and transformation.

Applied data warehousing methodologies, including Slowly Changing Dimensions (SCD), Change Data Capture (CDC), surrogate key assignment, and data cleansing, improving historical data tracking in Star Schema models.

Implemented AWS Glue Data Catalog for establishing a centralized metadata repository, improving data governance, asset discovery, and lineage tracking.

Integrated Collibra Data Governance for enforcing consistent data standards, compliance, and regulatory policies, ensuring organizational data integrity.

Enhanced data lifecycle management by designing data retention, archival, and purging strategies for compliance with GDPR, HIPAA, and enterprise governance policies.

Provided technical leadership and mentorship to junior engineers and cross-functional teams, conducting training sessions on best practices in data engineering, cloud architecture, and big data processing.

Designed and implemented CI/CD pipelines using AWS Code Pipeline, Terraform, and GitHub Actions, automating deployment of data pipelines, infrastructure, and monitoring alerts.

Experienced in developing interactive dashboards and reports using Tableau, enabling data-driven insights and business intelligence.

Experienced in working within Agile (Scrum, Kanban) and Waterfall methodologies, actively participating in sprint planning, daily stand-ups, backlog grooming, and project lifecycle management for efficient software delivery.

Technical Skills:

Aws Services: AWS Glue, Amazon Kinesis, Amazon S3, Amazon DynamoDB, AWS Identity and Access Management (IAM), Amazon S3 Glacier, AWS Lake Formation, Amazon Redshift, AWS Lambda, Amazon Managed Streaming for Apache Kafka (MSK), AWS Glue Data Brew, AWS Data Pipeline, AWS CloudWatch, AWS Step Functions, AWS Code Pipeline, Apache Airflow, Unity Catalog, Amazon Sage Maker, Step Functions.

Tool & Technologies: HDFS, MapReduce, YARN, Hive, Pig, Scala, Spark Streaming, Splunk, Sqoop, Zookeeper, Apache Spark, Apache Flink, Apache Kafka, MapR Control Systems, Oozie, IBM DataStage, IBM Infosphere, Tivoli Workload Scheduler, IIB, Web Services (SOAP/REST), WSDL, XSLT, JMS, and Shell Scripting.

ETL Tools: AWS Glue, AWS Data Pipeline, Informatica PowerCenter, IBM Information Server, Talend, Snowflake, Star Schema, Collibra, Hackolade, DAX, Airflow, Oozie and Tivoli.

Methodologies: Agile, Scrum, Waterfall, Kanban.

Languages: PySpark, SQL, PL/SQL, Python, PIG, Java, HiveQL, Scala.

Databases: Amazon RDS (MySQL, PostgreSQL, Oracle, MS SQL Server), Amazon Redshift, Teradata, HBase, MongoDB, Snowflake, Amazon DynamoDB, Cassandra.

Hadoop Distribution: Hadoop Distribution Cloudera, Horton Works.

Operating Systems: Windows, UBUNTU, LINUX, UNIX.

Version Control: AWS Code Commit, GitLab, GitHub, Bitbucket, VisualSVN, Assembla.

IDE & Build Tools Design: Eclipse, Visual Studio, Sublime, Databricks, Jupyter Notebook, GraphQL, TypeScript.

Education:

Master of Science Missouri State University, Springfield, Missouri
May 2014

Bachelor of computer Science Lovely Professional University, Jalandhar, Punjab
May 2012

Work Experience:

Client: Citi Bank, NJ

Apr 2022 - Present

Role: Sr. Data Engineer

Responsibilities:

Designed and implemented robust ETL pipelines using PySpark, Pandas, and Databricks, enabling ingestion and transformation of high-volume financial datasets from structured and semi-structured sources.

Led the development of batch and near real-time data ingestion frameworks using Spark Structured Streaming and Delta Lake, integrated with Azure Data Lake Storage.

Collaborated closely with data scientists, compliance teams, and product owners to gather data requirements and convert them into scalable data engineering solutions.

Applied advanced Spark performance tuning techniques—including partitioning, broadcast joins, and caching—achieving up to 35% improvement in job execution time.

Managed infrastructure deployment and automation using Terraform, GitHub Actions, and Azure DevOps, streamlining CI/CD for data pipelines and infrastructure provisioning.

Designed reusable, metadata-driven pipelines in Azure Data Factory and Apache Airflow for reliable orchestration of end-to-end workflows.

Integrated data quality validation layers using PySpark and SQL logic to ensure consistency, accuracy, and completeness of ingested data.

Ensured compliance with GDPR and financial data privacy regulations through role-based access control, encryption policies, and audit logging using Azure Purview and Databricks Unity Catalog.

Explored and piloted alternative orchestration tools like Dagster and Prefect for lineage tracking and advanced pipeline observability.

Delivered self-service analytics dashboards by integrating Snowflake, Power BI, and Redshift with curated and validated data layers.

Environment: Azure Data Factory, Azure Synapse Analytics, Azure Data Lake Storage Gen2 (ADLS Gen2), Delta Lake, Blob Storage, Parquet, Azure Databricks, JSON, Avro, Azure SQL Database, Change Data Capture (CDC), Power BI, Apache Airflow, PySpark, Materialized Views, Query Caching, Azure Monitor, Azure NSGs, Load Balancers, Autoscaling, Microsoft Purview, Azure RBAC, Active Directory (AD), Azure Key Vault, Snow Flake, Azure Log Analytics, Tableau, Agile, Airflow, Oozie and Tivoli.

Client: TennCare, State of Tennessee, TN.

Jun 2019 – Mar 2022

Role: Data Engineer

Responsibilities:

Spearheaded the design of end-to-end data pipelines in AWS Glue, Spark, and Databricks, supporting real-time ingestion from healthcare claims systems and external reporting agencies.

Built scalable data workflows using PySpark and Pandas for eligibility validation, provider mapping, and member demographics tracking across multiple systems.

Automated pipeline deployments using CI/CD pipelines with Jenkins, GitHub Actions, and CloudFormation, reducing manual errors and improving delivery velocity.

Collaborated with business analysts, Medicaid data teams, and healthcare data scientists to gather requirements for population health metrics and predictive analytics use cases.

Conducted deep optimization of Spark jobs and queries on EMR clusters, reducing execution time for batch jobs by up to 40%.

Designed unified logging and monitoring solutions using CloudWatch, Prometheus, and custom Python utilities for pipeline health and anomaly detection.

Ensured HIPAA compliance by implementing secure data masking, encryption at rest/in transit, and audit trails across the platform using AWS KMS and IAM policies.

Evaluated and introduced Prefect as a proof-of-concept for decoupling pipeline orchestration from code logic, improving traceability and retry handling.

Delivered curated data views and lineage reports using Tableau, Power BI, and AWS Redshift, improving insights for program operations and Medicaid audits.

Standardized version control and workflow across the engineering team using Git, branching strategies, and integrated Agile lifecycle tools like Jira.

Environment: Amazon S3, AWS Glue, AWS Data Pipeline, AWS Lake Formation, Amazon Redshift, Amazon RDS, AWS Lambda, Amazon EMR, Snowflake, Apache Spark, PySpark, Scala, Snowflake Time Travel, Streams & Tasks, Hadoop, Hive, Amazon MSK (Kafka), Apache Airflow, Airflow DAGs, AWS IAM, AWS VPC, AWS Cost Management, AWS CloudFormation, AWS Snowball, AWS Backup, AWS Disaster Recovery, Jenkins, CI/CD, JIRA, Airflow, Oozie, Tivoli, Git, Snowflake, Agile Methodologies.

Client: T Rowe Price Group Inc, MD.

Jan 2018 – May 2019

Role: Big Data Engineer

Responsibilities:

Designed and implemented scalable data ingestion pipelines using Apache Spark, Apache Kafka, and Apache Flink to collect structured and unstructured data from customer databases, IoT devices, and third-party sources.

Configured and administered Informatica Axon and Enterprise Data Catalog (EDC) for metadata discovery, glossary management, and mapping business terms to technical data assets.

Developed and maintained workflows in Informatica Data Quality (IDQ) for profiling, standardization, validation, and remediation of enterprise datasets.

Designed and implemented REST APIs for seamless integration between Informatica tools (Axon, EDC, IDQ) and external metadata repositories and reporting platforms.

Performed Informatica on-premises and hybrid administration tasks including installation, configuration, patching, upgrades, and backup strategies on Linux-based environments.

Implemented automated source-to-target data lineage mapping using advanced EDC scanners to support auditability, compliance, and data traceability across systems.

Applied DAMA DMBOK-aligned data governance practices to design scalable frameworks for data ownership, stewardship, and quality monitoring.

Configured Single Sign-On (SSO) authentication and integrated Informatica applications with enterprise identity management systems for secure access and governance.

Designed and consumed data payloads in JSON and XML formats for API-based data exchange and integration workflows.

Created operational and business dashboards in Tableau and Power BI to visualize data quality metrics, lineage reports, and governance KPIs extracted from Informatica.

Implemented and maintained containerized services using Docker and orchestrated distributed ETL tasks with Kubernetes, ensuring high availability and scalability.

Integrated SaaS cloud databases hosted on AWS and Azure, including secure data sync to Snowflake from third-party APIs and web services.

Developed machine learning-based predictive models in Spark MLlib for risk assessment, customer churn analysis, fraud detection, and claims processing automation.

Created interactive dashboards and business intelligence reports using Amazon Quick Sight, presenting insights to stakeholders, including executives, underwriters, and claims analysts.

Defined and enforced data governance policies by implementing metadata management, lineage tracking, and data cataloging using AWS Glue and AWS Lake Formation.

Designed Airflow DAGs and Oozie workflows to orchestrate and automate ETL jobs, scheduling batch processing tasks and dependency management.

Optimized Spark SQL and Redshift queries by leveraging Adaptive Query Execution (AQE), Dynamic Partition Pruning, Materialized Views, and Bucketing, improving query response times by 50%.

Provisioned and managed Amazon EMR and Amazon Sage Maker clusters, ensuring seamless scalability and high availability for big data workloads.

Collaborated with healthcare analytics teams to ingest and transform CMS Fee Schedules into Snowflake, enabling cost benchmarking and claims-based analytics for payer reimbursement optimization.

I am currently preparing for the SnowPro Core Certification, with hands-on experience in Snowflake Streams, Tasks, and performance tuning for analytical workflows.

Built and optimized real-time streaming data pipelines using Spark Structured Streaming and Apache Kafka on AWS EMR for upstream ingestion and downstream consumption.

Developed metadata-driven PySpark pipelines for batch and real-time processing across Amazon S3, RDS, and DynamoDB, improving ingestion efficiency by 40%.

Designed scalable data lakes on Amazon S3, integrating upstream data from APIs and downstream consumer feeds via Glue and Lambda.

Tuned and orchestrated Spark jobs using Scala to process multi-terabyte event logs, with error handling and checkpointing for reliable stateful processing.

Delivered analytics dashboards using Tableau by transforming and staging datasets in Snowflake and Redshift.

Worked closely with data scientists, analysts, and business stakeholders to translate requirements into scalable data solutions, ensuring successful project execution and timely delivery.

Established multi-cloud and hybrid cloud big data solutions, integrating Amazon S3, HDFS, and Snowflake to enable seamless data sharing and interoperability.

Environment: Apache Spark, PySpark, Apache Kafka, Apache Flink, Amazon S3, Amazon Redshift, Snowflake, Amazon DynamoDB, AWS Lake Formation, Trino (Presto), Parquet, ORC, Avro, AWS Glue, Kafka Streams, Spark Structured Streaming, Amazon Quick Sight, AWS Glue Data Catalog, Apache Airflow, Oozie, Amazon EMR, Amazon Sage Maker, PostgreSQL, MySQL, SQL Server, Terraform, Kubernetes, Jenkins, GitHub Actions, AWS Code Pipeline, YARN, Informatica, Talend.

Client: Toyota Motors , Plano, TX

Jan 2017 – Dec 2017

Role: Big Data Developer

Responsibilities:

Designed and deployed an Enterprise Data Lake to support analytics, data processing, storage, and reporting of large-scale, rapidly evolving datasets.

Gained expertise in the full Software Development Life Cycle (SDLC), including business requirements analysis, programming, database design, data warehousing, and business intelligence, applying both Star and Snowflake schema methodologies.

Developed and optimized T-SQL scripts, stored procedures, functions, table variables, indexes, common table expressions (CTE), views, and temporary tables in SQL Server for efficient query execution.

Led the design, development, and testing of ETL processes using Informatica PowerCenter for building robust data warehouse applications and test data management solutions.

Identified and addressed data quality issues at the field and column levels in source systems, improving data integrity through validation, cleansing, and error handling in ETL workflows.

Designed and developed ETL packages in SQL Server Integration Services (SSIS) to extract, transform, and load data from multiple sources, including SQL Server, Oracle, flat files, CSV, and XML.

Applied strong expertise in OLTP and OLAP data warehousing, data mining, data governance, and overall data management services, ensuring data quality and consistency.

Analyzed transactional database schemas and designed optimized star schema models to support business intelligence and reporting requirements.

Deployed SSIS packages into production environments, leveraging various package configurations for seamless migration and maintaining an environment-independent setup.

Developed and optimized SQL and PL/SQL scripts for database triggers, performance tuning, and efficient data processing.

Created complex T-SQL stored procedures, functions, triggers, cursors, and schema objects, enhancing database functionalities.

Designed and implemented Drill-through, Drill-down, Sub-reports, and Linked reports using SQL Server Reporting Services (SSRS) for advanced business insights.

Optimized SQL queries and enhanced SSIS package performance to improve report execution times and database consistency.

Applied Agile methodology using the Scrum framework, ensuring efficient project management and development through iterative sprints and continuous improvements.

Environment: SQL Server, T-SQL, PL/SQL, Informatica PowerCenter, SSIS, SSRS, OLTP, OLAP, Star Schema, Snowflake Schema, Data Warehousing, Data Mining, Data Governance, ETL, Stored Procedures, Functions, Indexing, Common Table Expressions (CTE), Views, Temporary Tables, Oracle, Flat Files, CSV, XML, Database Triggers, Performance Tuning, Drill-through Reports, Drill-down Reports, Sub-reports, Linked Reports, Agile, Scrum.

Client: American Airlines, Fort Worth, Tx

Jan 2016 - Dec 2016

Role: Hadoop Developer

Responsibilities:

Developed and maintained data ingestion pipelines using Apache Flume, Apache Kafka, and Apache Sqoop to import structured and unstructured data from relational and NoSQL databases into HDFS.

Designed and optimized ETL workflows using Apache NiFi and Apache Airflow to automate data movement across distributed systems.

Managed large-scale datasets in HDFS, optimizing replication, compression (Snappy, LZO, Gzip), and fault tolerance for efficient data retrieval.

Developed MapReduce programs and Spark (PySpark/Scala) applications to process large datasets, leveraging Data Frames, RDDs, and partitioning for performance optimization.

Built and optimized Hive queries using bucketing, partitioning, and indexing to reduce query latency and improve analytical performance.

Integrated Apache Tez to enhance Hive query execution and performance.

Developed real-time streaming solutions using Apache Kafka, Apache Flink, and Spark Streaming for processing IoT, log, and social media data.

Designed and managed workflow automation using Apache Oozie and Apache Airflow for scheduling ETL jobs and monitoring dependencies.

Integrated Hadoop with HBase and Cassandra, designing row-key and column-family structures for low-latency data retrieval.

Implemented data security best practices using Apache Ranger for access control and Kerberos authentication, ensuring compliance with governance policies.

Monitored Hadoop cluster performance using YARN, Ambari, and Cloudera Manager, tuning Spark jobs by optimizing memory allocation and shuffle partitions.

Worked in Agile Scrum environments, collaborating with data engineers, analysts, and business teams to define data requirements and optimize processing workflows.

Environment: Hadoop, HDFS, Apache Spark, PySpark, Scala, Hive, Impala, Apache Flink, Apache Kafka, Apache Sqoop, Apache Flume, Apache NiFi, MapReduce, HBase, Cassandra, MongoDB, Apache Oozie, Apache Airflow, Apache Ranger, Apache Atlas, Apache Tez, YARN, Ambari, Cloudera Manager, Kerberos, Git, Jenkins, Prometheus, SQL, Python.

Client: Morgan Stanley, Chicago, IL

Jun 2014 – Dec 2015

Role: Data Warehouse Developer

Responsibilities:

Created SQL Mail Agent jobs, alerts, and scheduled DTS/SSIS packages to automate recurring data processing workflows.

Managed and updated Erwin models for logical and physical data modeling of Consolidated Data Store (CDS), Actuarial Data Mart (ADM), and Reference DB to align with evolving user requirements. Utilized TFS for version control and tracking environment-specific script deployments, ensuring consistency across database environments.

Exported current data models from Erwin to PDF format and published them on SharePoint for seamless user access and collaboration.

Developed, administered, and maintained databases, including Consolidated Data Store, Reference Database, and Actuarial Data Mart, to support enterprise data needs.

Wrote and optimized triggers, stored procedures, and functions using Transact-SQL (T-SQL), ensuring efficient database operations and query performance.

Deployed scripts across multiple environments following Configuration Management and Playbook guidelines for controlled database updates.

Created and managed files and file groups, optimizing table and index structures to enhance query performance and database efficiency.

Tracked, analyzed, and resolved database-related defects using Quality Center for effective issue management and resolution.

Maintained and administered database security by managing users, roles, and permissions within SQL Server to enforce access control policies.

Environment: SQL Server 2008/2012 Enterprise Edition, SSRS, SSIS, T-SQL, Windows Server 2003, Performance Point Server 2007, Oracle 10g, Visual Studio 2010.